

Chapter #2: Basic physics of Semi-Conductors

Lecture #2:

→ Elec circuits are based on complex semi-conductor structure.

• need to study:

↳ movement of charges inside circuit components

→ develop circuit models of devices.

→ solid-state devices

Semi-Conducting material & their properties:-

- periodic table

III	IV	V
Boron	C	P
Al	Si ¹⁴	As
Ga	Ge	

IV: semi-conductors

Si: Silicon base semi-conductors

↳ abundant, easy processing

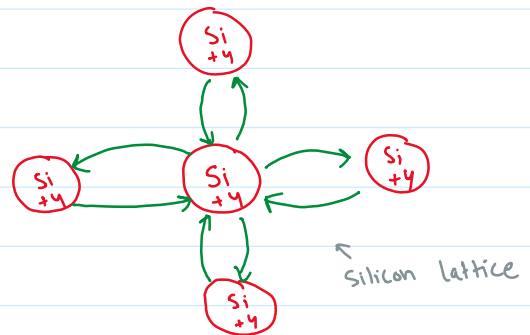
Eg of silicon = 1.12 eV

Silicon Lattice:

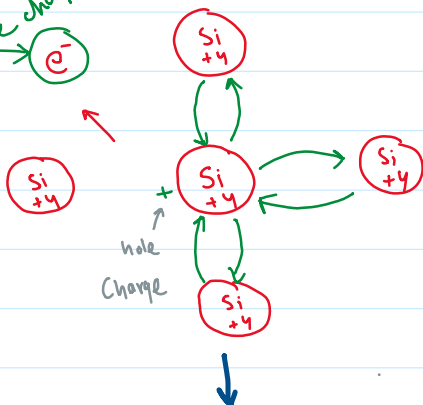
→ regular lattice

→ each atom is surrounded by 4 atoms

→ covalently bonded.



Mobile charge, carrier, free charge e^-



At room temperature
(27°C → 300K)

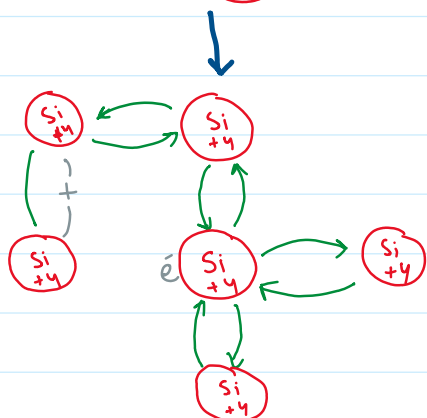
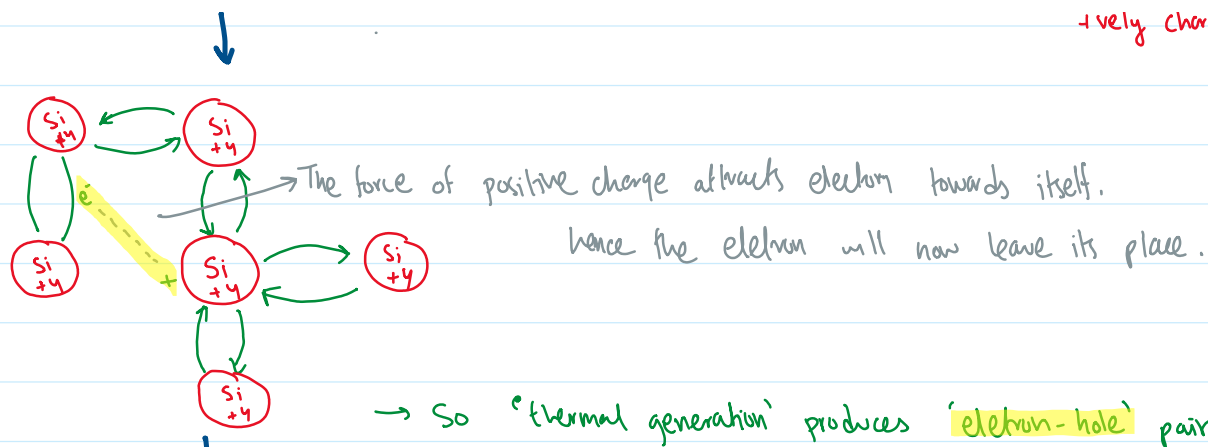
• bond breaks → a free e^- is available

when free e^- leaves parent atom

creates a hole behind

↓
+vely charged

positively charged



→ -ve charge moves this way
← +ve charge moves the other way.

Band-gap energy:

$$n = p = n_i = BT^{3/2} e^{-E_g/2kT}$$

where:

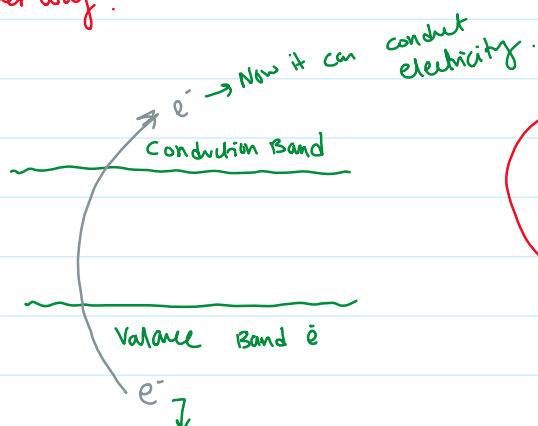
E_g = Band-gap (material dependent)

n = concentration of e^-

p = concentration of holes

n_i = intrinsic conc.

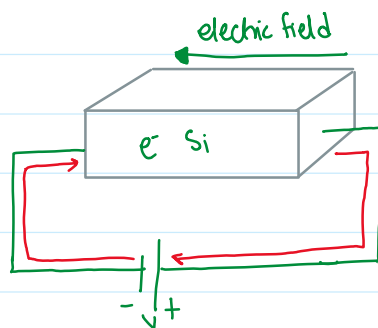
B = material constant



Example 2.1:

$$n_i(300K) = 1 \times 10^{10} \text{ e/cm}^3 \text{ (Ans)}$$

$Cu \rightarrow 10^{18} \text{ e/cm}^3$ - as compared to copper
silicon seems to be an insulator.



$\text{Cu} \rightarrow 10^{18} \text{ e/cm}^3$ - as compared to copper
silicon seems to be an insulator.



• Modification of Carrier Densities:

→ Doped/Extrinsic Si

↳ Controlled addition of impurity atoms in Si

• Lecture #3

- Doped semi-conductors

↳ controlled addition of other atoms. (P)

$n_n \approx N_D$ ← Conc. of donors.
Conc. of e^- ↑
in the n-type

$$n_i(\text{Cu}) \approx 10^{18-20} \text{ e/cm}^3$$

$$n \cdot p = n_i^2$$

$$\frac{n_n}{N_D} \cdot p_n = n_i^2$$

$$p_n = \frac{n_i^2}{N_D} \rightarrow \frac{10^{20}}{10^{16}}$$

$$n_n = 10^4 \text{ h/cm}^3$$

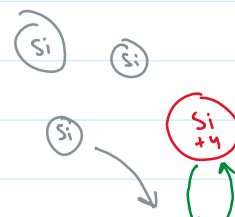
- as we increase Phosphorus, conc. the conc. of holes decreases, because phosphorus replaces silicon that creates holes.

e^- in n-type → majority carriers.

holes in n-type → minority carriers.

• To create more holes: (P-type semi-conductors)

- as Silicon moves towards Boron, so it creates holes



∴ Boron becomes an acceptor Atom.

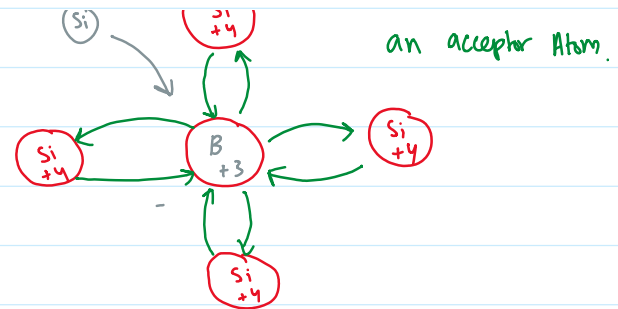
- as Silicon moves towards Boron, so it creates holes.

- greater the boron conc. greater the holes.

So,

$$p_p \approx N_A$$

$$n_p = \frac{n_i^2}{N_A}$$



So,

n-type

p-type

Independent of temperature.

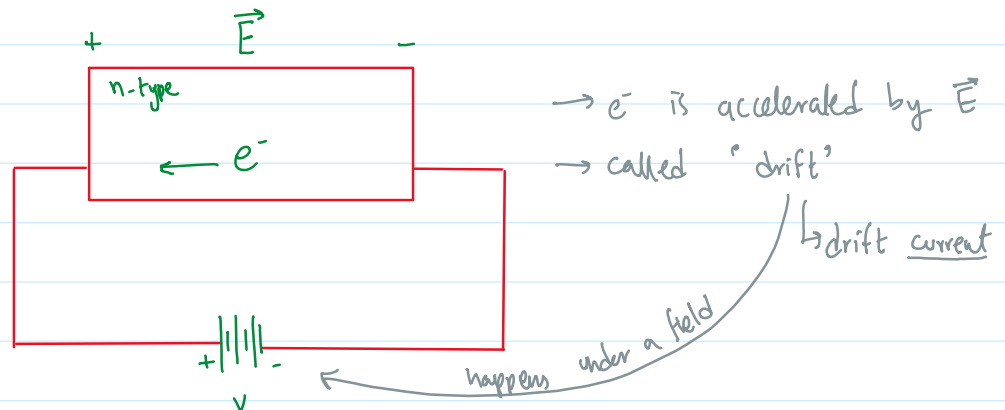
$$n_n \approx N_D$$

$$p_n \approx \frac{n_i^2}{N_D}$$

$$p_p \approx N_A$$

$$n_p \approx \frac{n_i^2}{N_A}$$

Transport of Carriers inside Doped-Si:



drift velocity (m/s)
 $v \propto E$

$$v = \mu E$$

mobility of e^- in Si

mean free path:- how much electron can travel without colliding the lattice
 $\rightarrow$ Distance.

$$\mu_n = 1350 \text{ cm}^2/\text{V}\cdot\text{s}$$

$$\mu_p = 480 \text{ cm}^2/\text{V}\cdot\text{s}$$

SS: Why holes are heavier/slower than electrons.

$$\mu_p = 480 \text{ cm}^2/\text{V.s}$$

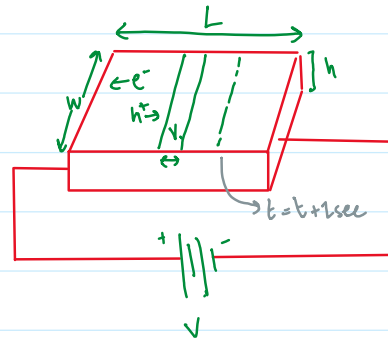
→ Charge passing through cross-sectional:- (in 1 second)

$$\boxed{\frac{V}{s} \cdot w \cdot h} \rightarrow \text{velocity}$$

→ if there are $n e^-$ βq is the charge of e^-

$$\boxed{I = -q n \cdot w \cdot h \cdot n}$$

∴ Now replace with drift velocity.



$$\boxed{J_n = \mu_n E n q}$$

↑ current density.

→

$$\boxed{J_{\text{total}} = \mu_n E n q + \mu_p E p q}$$

↑ electron density ↑ holes density

further solution:-

$$\boxed{J_{\text{total}} = q E (\mu_n n + \mu_p p)}$$

• example 3.3:-

Find ρ

a Intrinsic Si

b p-type Si → $N_A = 10^{16}/\text{cm}^3$

Use:-

$$n_i = 1.5 \times 10^{10}/\text{cm}^3$$

$$\mu_n = 1350 \text{ cm}^2/\text{V.s}$$

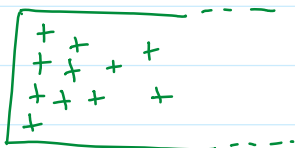
$$\mu_p = 480 \text{ cm}^2/\text{V.s}$$

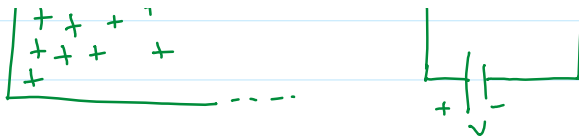
$$\therefore \sigma = 1/\rho$$

a) $\rho = 2 \times 10^5 \Omega \cdot \text{cm}$

b) $\rho = 1.56 \Omega \cdot \text{cm}$

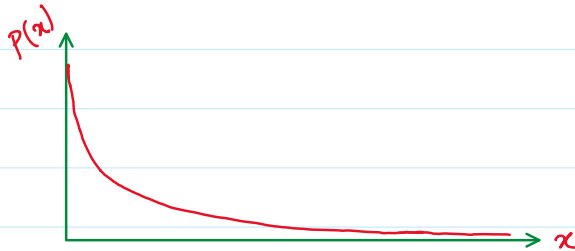
• Diffusion Current:-





→ Current diffuses from high concentration area to low concentration area, without any external force.

let $p(x)$ = Concentration of holes

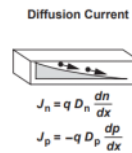
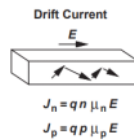


- for holes:

$$J_p \propto \frac{dp(x)}{dx}$$

$$J_p = -q D_p \frac{dp(x)}{dx}$$

↑
holes · diffusion density ↑
diffusion current.



- for electrons:

$$J_n = q D_n \frac{dn(x)}{dx}$$

↑
electron diffusion density.

7. Diffusion vs Drift (Very Important)	
Drift Current	Diffusion Current
Due to electric field	Due to concentration gradient
$J = q n \mu E$	$J = q D \frac{dn}{dx}$
Dominant under bias	Dominant at junction

- Total diffusion current density:

$$J_{total} = J_n + J_p$$

$$J_{total} = q \left\{ D_n \frac{dn(x)}{dx} - D_p \frac{dp(x)}{dx} \right\}$$

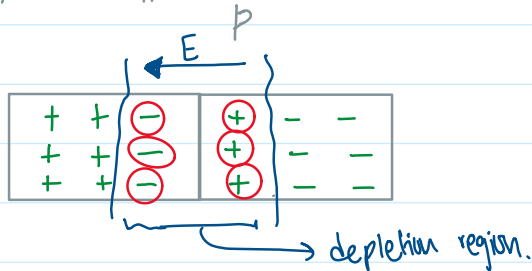
- Summary:
 $n_i = 10^{10} / \text{cm}^3$
n-type $n_n = 10^{18}$
p-type $p_p = 10^{18}$

- PN - Junction: — Junction

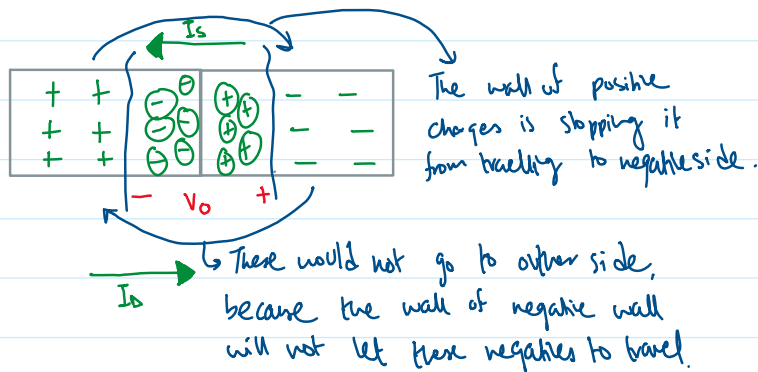
• PN - Junction:



- 1 $\hookrightarrow$ holes diffusing towards n
- $\hookrightarrow e^-$ " " " p

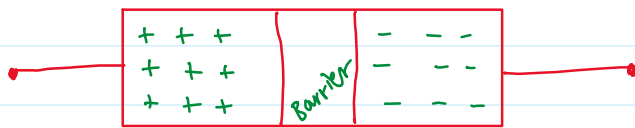


- 2 $\hookrightarrow$ holes & electron would recombine
3. immobile charge at junction gets exposed
 $\hookrightarrow$ given rise to a $V \propto E$ at the junction.



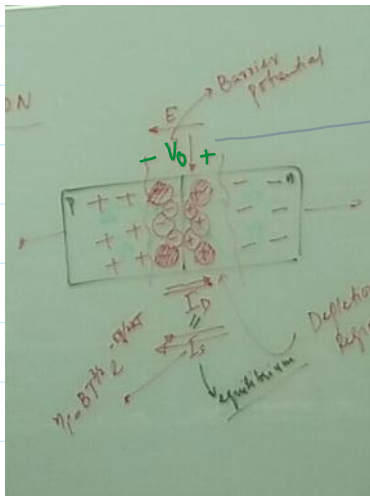
4. a depletion region is form at the junction
5. Junction Barrier potential stops further Diffusion V_o .
6. But, n_i due to thermal generation
 $\hookrightarrow$ it gives rise to drift current I_s .
and it balances the diffusion current.
7. equilibrium is obtained $I_o \approx I_s$

So, a barrier is observed:-



↑
if we overcome this by these two terminals
so it will cause electrical switching.

• Lecture #5



$$V_0 \propto \frac{N_D \cdot N_A}{n_i^2}$$

10^{18} (for N_D)
 10^{12} (for N_A)
 10^{20} (for n_i^2)

So,

$$V_0 = V_T (36.84)$$

makes 0.7V

Thermal voltage = $\frac{kT}{q}$

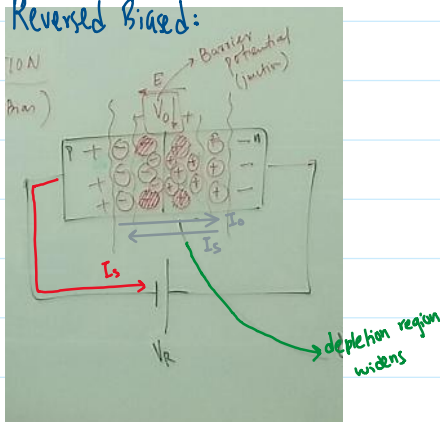
↓
at room temp

k — Boltzmann constant
 T — room temp
 q — charge

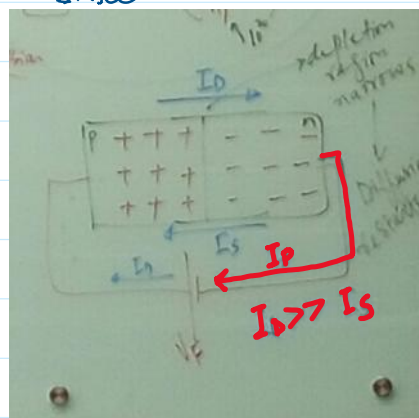
$$= \frac{1.38 \times 10^{-23} \times 300}{1.6 \times 10^{-19}} \approx 26 \text{ mV}$$

• Now under a bias point (external voltage):

Reversed Biased:



forward Biased:-



$$I_D = I_s \left[e^{V/V_T} - 1 \right]$$

↑ Diffusion current
↑ Drift current / Reverse saturation current

$$I_s = A q n_i^2 \left\{ \frac{D_n}{N_A L_n} + \frac{D_p}{N_D L_p} \right\}$$

Area of diode, Greater the diode, greater its thermal generation

Area $\approx 10^{-12} > 10^{-15}$ Typical range.

• Under **Reverse Biased**:-

$$I_D = I_s (e^{-V/V_T} - 1)$$

$$\therefore V_T = 26 \text{ mV}$$

let suppose $V_R = 100 \text{ mV} \rightarrow \text{very small}$

$$I_D \approx -I_s$$

← leakage current

• Under **forward Biased**

$$I_D = I_s (e^{V_F/V_T} - 1)$$

$$\therefore V_F = 0.7 \text{ V}$$

$$I_D = 10^{12} \cdot I_s$$

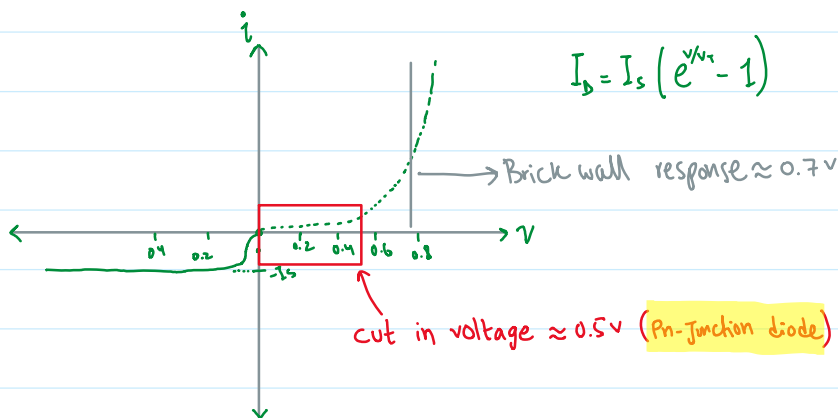
• diodes:



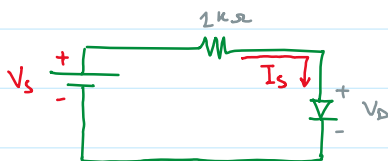
- Ideal diode:
 - doesn't conduct any current under Reverse biased
 - Short circuit under forward biased.

$$-V \quad + \quad i = 0$$

$$V = 0 \quad i = \infty$$



• Circuit analysis:



• Assume D_1 is forward biased.

1. ...

- Assume D_1 is forward biased.

$$\hookrightarrow V_s = V_R + V_D$$

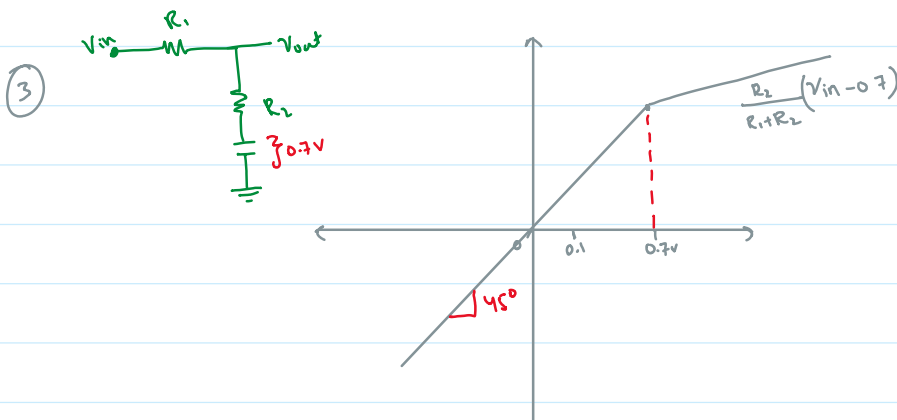
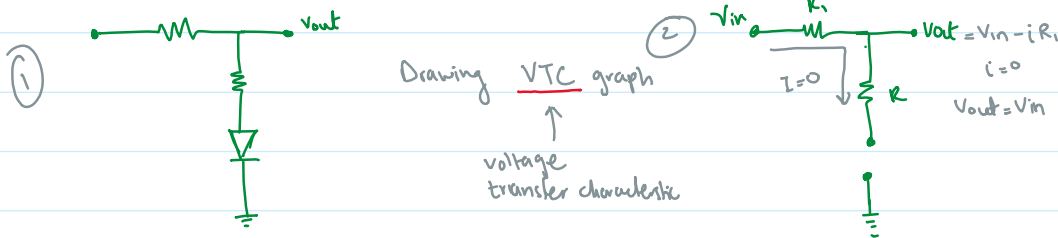
$$= I_s R + V_T \ln \left(\frac{I_x}{I_s} + 1 \right) \rightarrow \text{as it is a very complex equation, so we assume } 0.7V \text{ (forward biased)}$$

So,

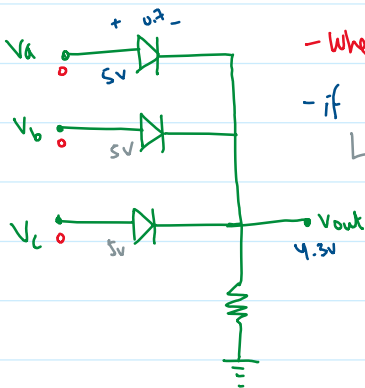
$$V_s = I_x \cdot 1k + 0.7$$

lets suppose $V_s = 5V$ Then $I_x = \frac{5 - 0.7}{1000} = 4.3mA$ → as direction of current is positive and answer is correct, so we can tell the assumption was correct.

- figure 3.14:



- Problem:



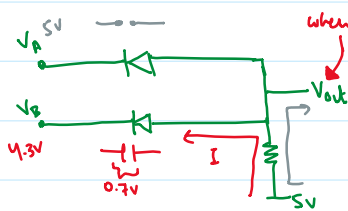
- when all three inputs are zero, output is zero.

- if 5V is applied to V_a , output is 4.3V

→ if any of them gets on, output also gets high.

its an OR-gate

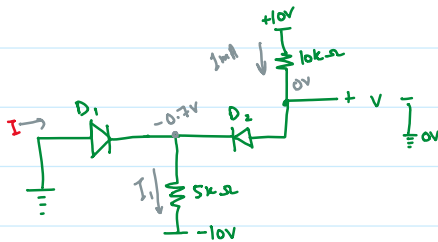
• Problem:



→ even if one of them is low (5V), the output is low.

• It's an AND gate

• Problem:-



• Region Assumption:

D_1 is on

D_2 is on

$$I = \frac{-0.7 - (-10)}{5k} = 1.86mA$$

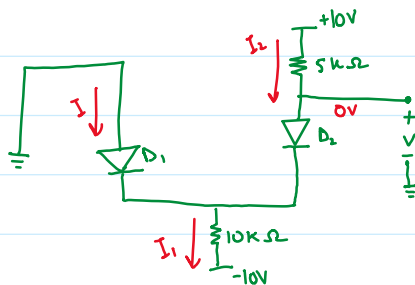
$$I = 2mA$$

$$I_1 = I + I_2$$

$$1.86mA = 2mA + I_2$$

$$0.86mA = I_2$$

• Problem:



$D_1, D_2 \rightarrow$ on

$$I_1 = \frac{-0.7 - (-10)}{10,000}$$

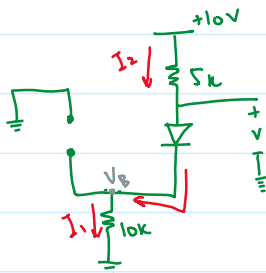
$$I_1 = 0.93mA$$

$$I_2 = \frac{10 - 0}{5k} = 2mA$$

Now, Consider only D_2 is on:-

$$I = I_1 - I_2 = -1.07mA$$

Inconsistent Answer
(with D_1 on)



So,

$$I_2 = \frac{10 - 0.7 - (-10)}{15k} = 1.28mA$$

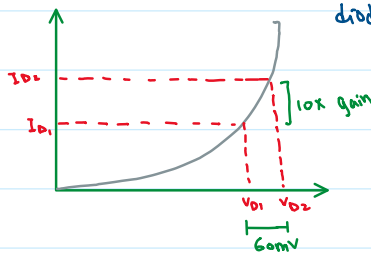
$$V_B = 2.8V$$

- Iterative Analysis using exponential Model: we used to assume brickwall at 0.7 V, but now we will calculate actual diode voltage.

$$I_D = I_S e^{v_D/V_T}$$

$$I_{D1} = I_S e^{v_{D1}/V_T} \quad \text{--- i}$$

$$I_{D2} = I_S e^{v_{D2}/V_T} \quad \text{--- ii}$$



$$\frac{I_{D2}}{I_{D1}} = \frac{I_S e^{v_{D2}/V_T}}{I_S e^{v_{D1}/V_T}}$$

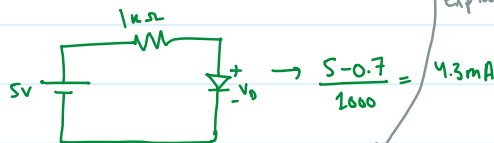
$$V_{D2} - V_{D1} = V_T \ln \left(\frac{I_{D2}}{I_{D1}} \right) = 2.3 V_T \log \left(\frac{I_{D2}}{I_{D1}} \right)$$

if $I_{D2} = 10 I_{D1}$, then, (at room temperature)

$$V_{D2} - V_{D1} = 2.3 V_T \approx 60 \text{ mV} < \text{per decade}$$

Problem:

$$I_D = 1 \text{ mA} @ 0.7 \text{ V}$$



explanation

if we want 10mA, so drop voltage will get 0.76 V
 if we want 100mA, so drop voltage will get 0.82 V (100mA)

So solving problem,

$$V_{D2} - V_{D1} = V_T \ln \left(\frac{I_{D2}}{I_{D1}} \right)$$

$\uparrow \quad \quad \uparrow$
 $? \quad \quad 0.7$
 $\quad \quad \quad \uparrow$
 $\quad \quad \quad 1 \text{ mA}$

So,

$$V_{D2} = 0.7 + V_T \ln \left(\frac{4.3 \text{ mA}}{1 \text{ mA}} \right)$$

$$\therefore V_T \ln = 0.06 \log$$

So,

$$V_{D2} = 0.7 + 0.06 \log \left(\frac{4.3}{1} \right) = 0.738 \text{ V} \rightarrow \text{first iteration.}$$

- 2nd iteration:-

$$\frac{5 - 0.738}{1000} = I_{D2} = 4.262 \text{ mA}$$

Now keep doing iterations to get closer to the realistic voltage.

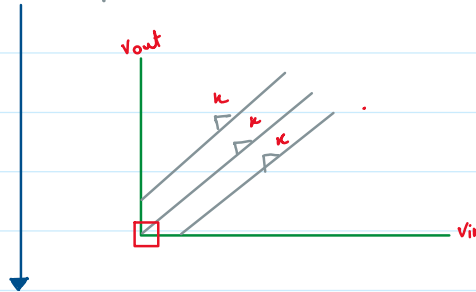
$$V_{D3} = V_{D2} + 0.06 \log \left(\frac{I_{D3}}{I_{D2}} \right) = 0.738 + 0.06 \log \left(\frac{4.262}{4.3} \right) = 0.7377 \text{ V}$$

Now keep doing iterations to get closer to the realistic voltage.

$$V_{03} = V_{02} + 0.06 \log\left(\frac{I_{03}}{I_{02}}\right) = 0.738 + 0.06 \log\left(\frac{4.262 \text{ mA}}{4.3 \text{ mA}}\right) = 0.7377 \text{ V}$$

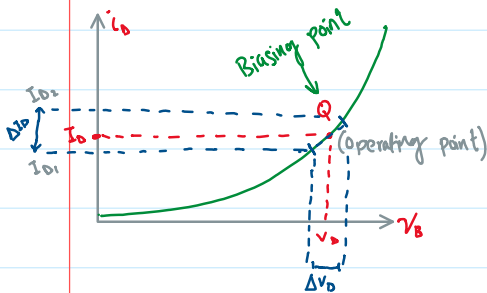
we stop here because the difference is in 3rd decimal.

- Large Signal Model ? Small signal Model



if a diode is made to experience small changes in $V \& I$
 $\rightarrow$ Non-linear device can be changed in 'piece wise' linear device

- Small-signal model:-



$$I_{D2} = I_S e^{\frac{(V_{D1} + \Delta V)}{V_T}} = I_S e^{\frac{V_{D1}}{V_T}} \cdot e^{\frac{\Delta V_D}{V_T}}$$

$$\therefore e^x = 1 + x + \frac{x^2}{2!} \dots$$

$$\text{so, } e^{\frac{\Delta V_D}{V_T}} \approx 1 + \frac{\Delta V_D}{V_T}$$

ignore rest because it's a very small value. $\therefore \frac{\Delta V_D}{V_T} \ll 1$

$$\frac{\Delta I_D}{\Delta V_D} \text{ slope of 'Q'}$$

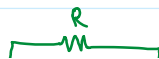
so,

$$I_{D2} = \underbrace{I_S e^{\frac{V_{D1}}{V_T}}}_{I_{D1}} + \underbrace{\frac{\Delta V_D}{V_T} I_S e^{\frac{V_{D1}}{V_T}}}_{\Delta I_D}$$

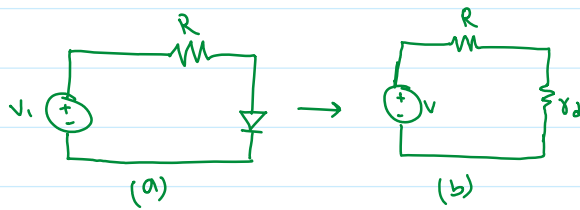
$$\Delta I_D = \frac{\Delta V_D}{V_T} I_{D1}$$

$$\frac{\Delta V_D}{\Delta I_D} = \frac{V_T}{I_D} = r_d \leftarrow \text{incremental resistance}$$

- so, we basically converted circuit (a) into circuit (b):



(... for small signals)



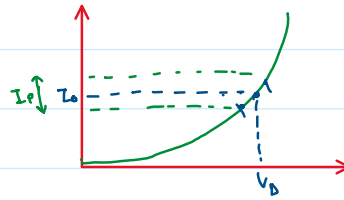
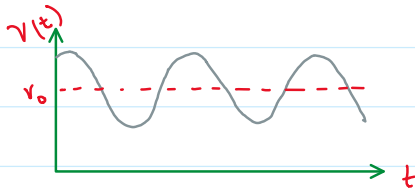
(only for small signals)

• Example 3.19: (DC) Q

$v(t) = V_0 + \boxed{V_p \cos \omega t}$ ← small signal (AC)

if $v(t)$ is applied across a diode: if $V_p < V_T$

find $I_D(t)$.



Solution:

$$\therefore r_d = \frac{V_T}{I_0}$$

$$\therefore I_0 = I_S e^{V_0/V_T}$$

$$I_p = \frac{V_p}{r_d} = \frac{I_0 V_p}{V_T}$$

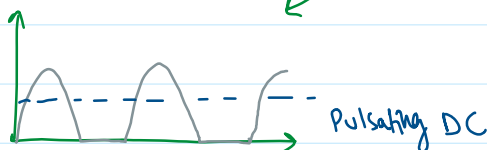
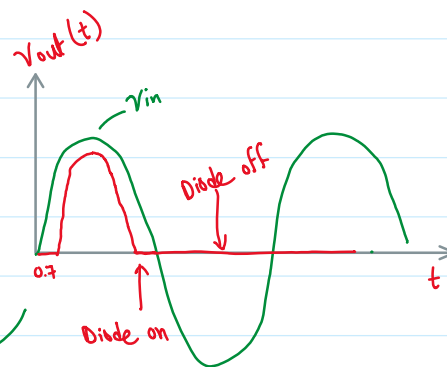
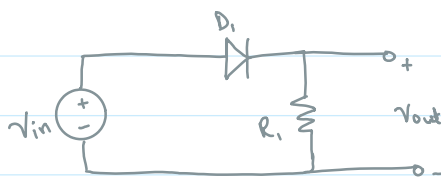
$$I_D(t) = I_0 + I_p$$

$$I_D(t) = I_S e^{v(t)/V_T} + \frac{I_0 V_p}{V_T}$$

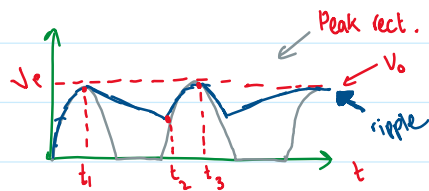
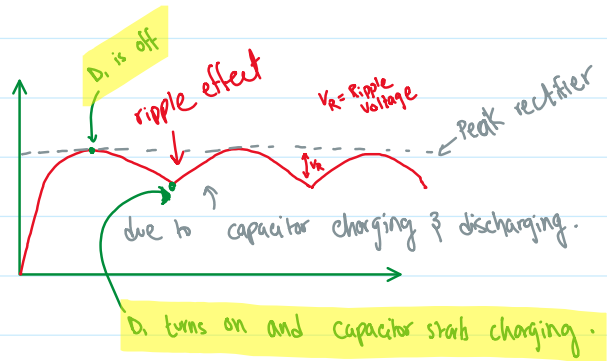
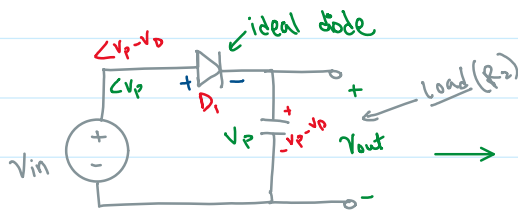
$$I_D(t) = I_S e^{V_0/V_T} + \frac{I_0 V_0 \cos(\omega t)}{V_T}$$

• Application of Diodes:-

- Half-wave rectifier:



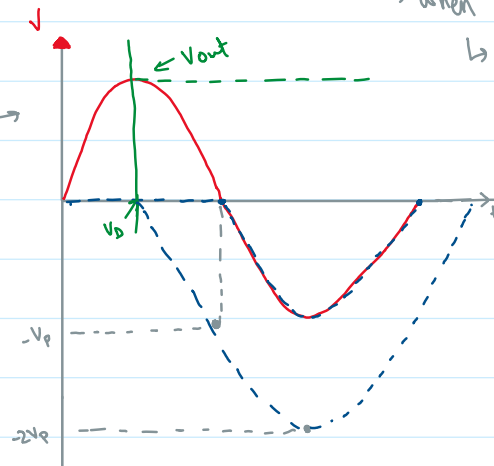
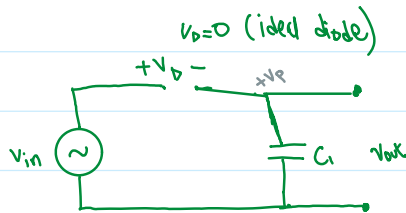
- Peak-wave Rectifier:



- V_{out} starts falling after t_1
- R_L provides discharging path
- During the next +ve cycle

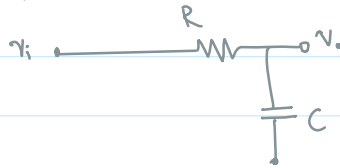
↳ when $V_{in} > V_{out}$
↳ C_{in} is charging again

Ex 3.24:



Now,

• Now,



$$V_o = V_i (1 - e^{-t/RC})$$

$RC \rightarrow$ time constant (τ)

at t_2 :

$$V_p - V_R = V_p e^{-T/RC}$$

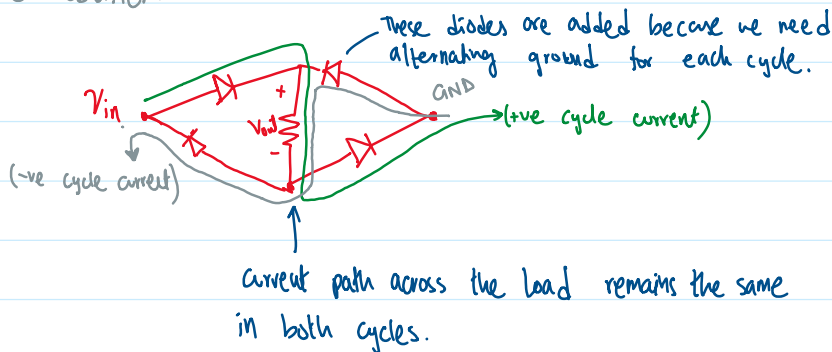
$$V_p (1 - e^{-T/RC}) = V_R \quad (T \rightarrow \text{time period of charging discharging})$$

$RC \gg T$: for a practically small V_R

$$V_R = V_p \left(\frac{T}{RC} \right)$$

$\uparrow$ Time
 $\uparrow$ load resistance
 $\uparrow$ V_{ripple}
 $\uparrow$ V_{peak}
 $\uparrow$ Capacitance

• Full wave rectifier:



Now,

